ISHA KC

ishakc432@gmail.com +977 9803429208 — +91 9740967386

Aerospace Engineering graduate with interest and project-based experience in experimental aerodynamics, CFD, designing and systems studies. Currently an Avionics and Airframe Systems Technical Services Engineer at Airbus India, working on aircraft system-level analysis on Airbus fleet.

Education

Pulchowk Engineering Campus, IOE *2021 – 2025*
Bachelor in Aerospace Engineering *79.72%*

Technical Tool skills

Modelling & Simulation: ANSYS Fluent, OpenFOAM, XFLR5, X-Plane, PlaneMaker, MATLAB Simulink
Programming & Documentation: MATLAB, C, Python, LaTeX
CAD & Design: CATIA, Solidworks, OpenVSP
Aviation Tools : MIS (AMASIS, AMOS), SKYWISE

Experience

Avionics and Airframe Systems Technical Services Engineer – Airbus India *Feb 2026 – Present*

- Analyse avionics and aircraft systems technical documentation to assess its impact on the aircraft as a whole and on the associated systems, interfaces, and dependencies.
- Perform reliability analysis of faults across avionics and aircraft systems, identifying recurring patterns and trends to support evidence-based engineering decisions.
- Set up and structure aircraft systems in maintenance information systems, accounting for system interfaces, assemblies, and configuration relationships.
- Assess applicability and consistency of technical data across aircraft configurations to maintain system-level integrity and traceability.
- Coordinate with systems, configuration, and engineering teams to resolve technical queries and support informed decision-making.

Projects

Oscillating-Vane Gust Generator Development (1-Cosine Profile)

- Designed and developed an oscillating-vane gust generator to produce controlled sinusoidal motion and approximate 1-cosine longitudinal gust profiles.
- Developed MATLAB-based analytical models of vane motion and gust generation, applying theoretical formulations to characterize the generated gust profile.
- Implemented Arduino-based actuation and control, integrating mechanical design, motion control, and experimental testing.
- Conducted unsteady CFD simulations in ANSYS Fluent to investigate the transient flow field generated by the oscillating vanes.
- Performed experimental flow visualization using smoke and high-speed imaging to examine gust development and evaluate generator performance.

Low-Speed Wind Tunnel Design and Fabrication

- Designed and fabricated a low-speed wind tunnel as an integrated experimental system.
- Optimized contraction ratio, test section, and flow conditioning for uniform velocity profiles.
- Validated system performance through measurement and comparative analysis.

Preliminary Design and Systems Integration of Point-to-Point Cargo UAV

- Led system-level trade studies integrating payload, propulsion, aerodynamics, and mission requirements.
- Performed mission-based performance analysis, stability assessment, and subsystem sizing.
- Evaluated interfaces between structural layout, aerodynamic configuration, and manufacturability.
- Presented the integrated system architecture and performance outcomes in a technical poster.

CFD-Based Multiphysics Studies Using OpenFOAM

- Simulated natural convection in a closed cavity to study gravity and thermal boundary effects.
- Modelled stirred tank flow using MRF and k–ε turbulence model.

- Conducted mesh sensitivity and parametric studies, and validated results against benchmark data.

1U CubeSat Systems Study, Structural Design and Assembly

- Carried out a system-level study and physical assembly of a 1U CubeSat (ORION Space training), focusing on structural design.
- Integrated EPS, OBC, and communication subsystems structurally, working within interface and packaging constraints.
- Demonstrated the CubeSat model at Engineering Expo 2024.

Design and Manufacturing of Small-Scale Aircraft

- Integrated aerodynamic, structural, and control aspects in the design of a small-scale aircraft.
- Developed detailed CAD models for CNC-based manufacturing; assessed static stability and configuration feasibility.

Internship

Fleet Technical Management Intern – Airbus India *August 2025 - Feb 2026*

- Supported Maintenance Program and Configuration Management activities for in-service aircraft.
- Analyzed technical publications to support fleet-level decision-making for Airbus operators and airlines.
- Performed reliability and operational delay analysis for Airbus fleets.
- Supported Entry Into Service activities, including CMR/ARC compliance and lease-related documentation.
- Assisted with engine oil consumption monitoring and supported engine, APU, and landing gear transactions.
- Managed technical library operations, including collection and assignment of technical publications.
- Supported continuing airworthiness activities in line with EASA Part-M ensuring maintenance data and records aligned with regulatory expectations.

Achievements & Leadership

- Participated in **IIT FOSSEE Summer Fellowship 2025** (OpenFOAM).
- Awarded multiple academic scholarships and the Best Student Award for academic and extracurricular performance.
- 3rd Place, All-Nepal Debate Competition (Physics Initiatives Nepal).
- Scout Troop Leader for 2 years; led a 50-member adventure trail team.
- Literary Coordinator, SEDS Pulchowk.

Additional Activities

- Active member of Initiative for Girls in Physics (IGP), SEDS(Students for the Exploration and Development of Space) Pulchowk, and Physics Initiatives Nepal.
- Completed 25+ hours of community service and technical outreach; completed a 7-day ANSYS workshop on simulation workflows.